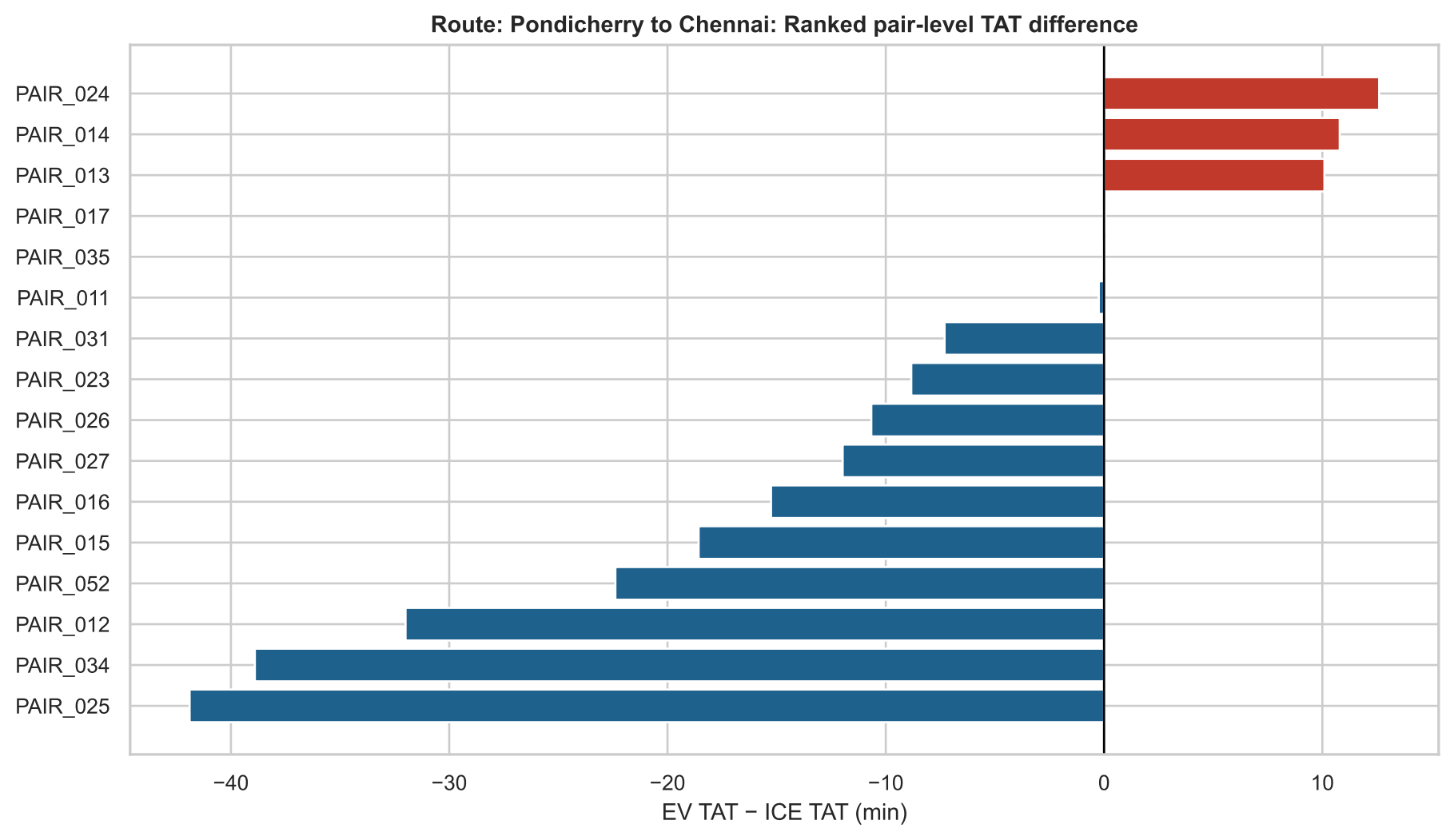
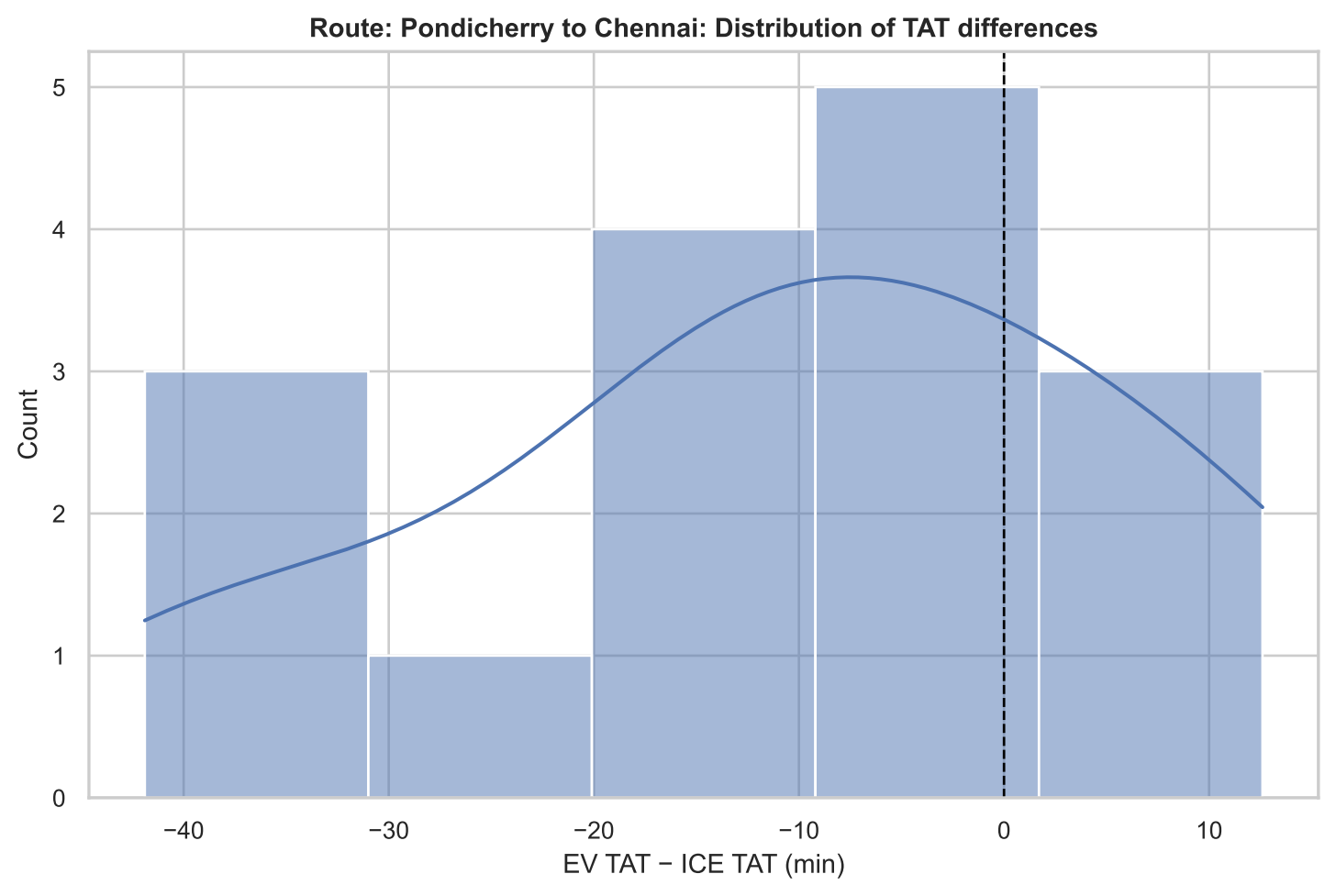
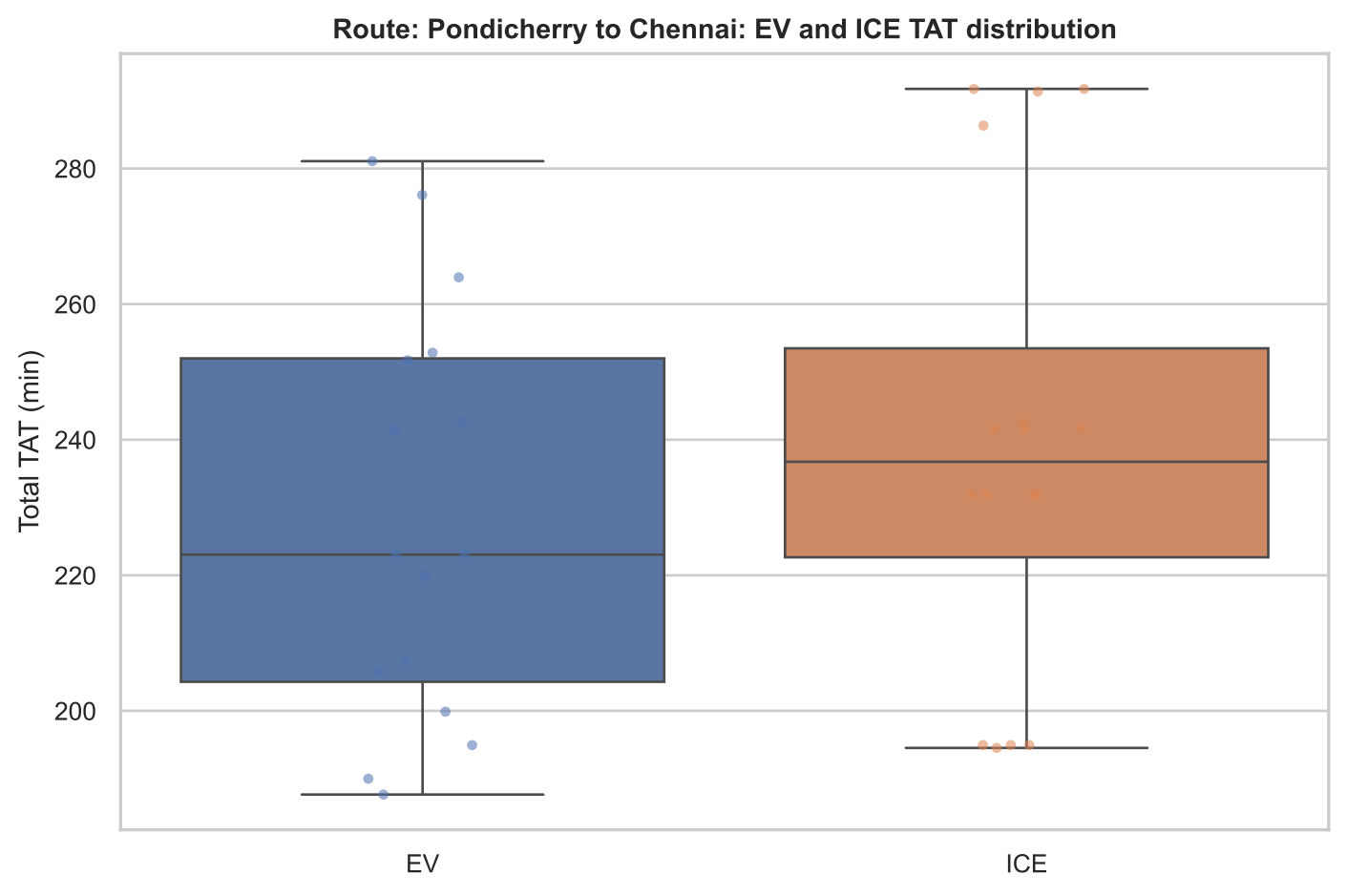
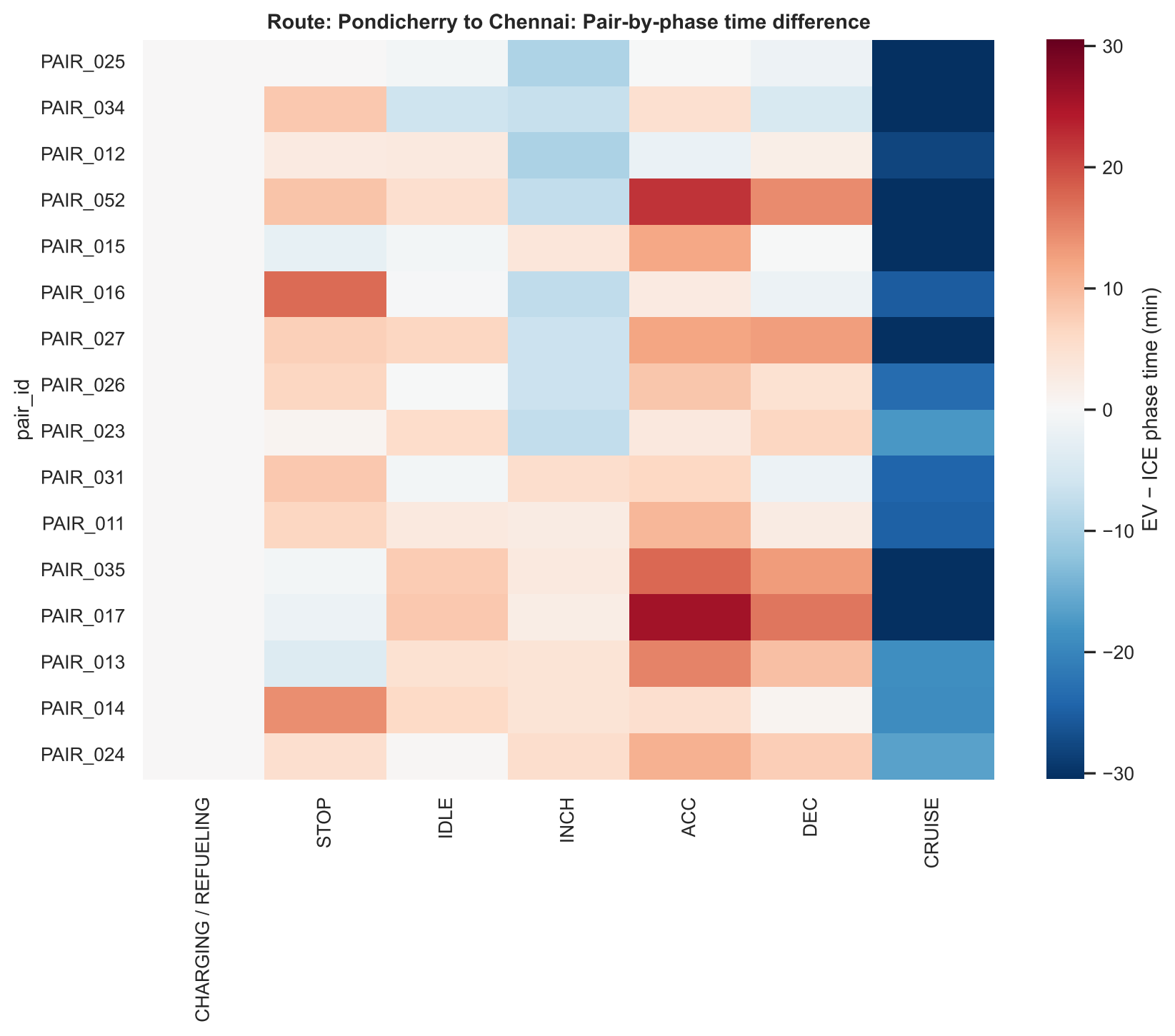
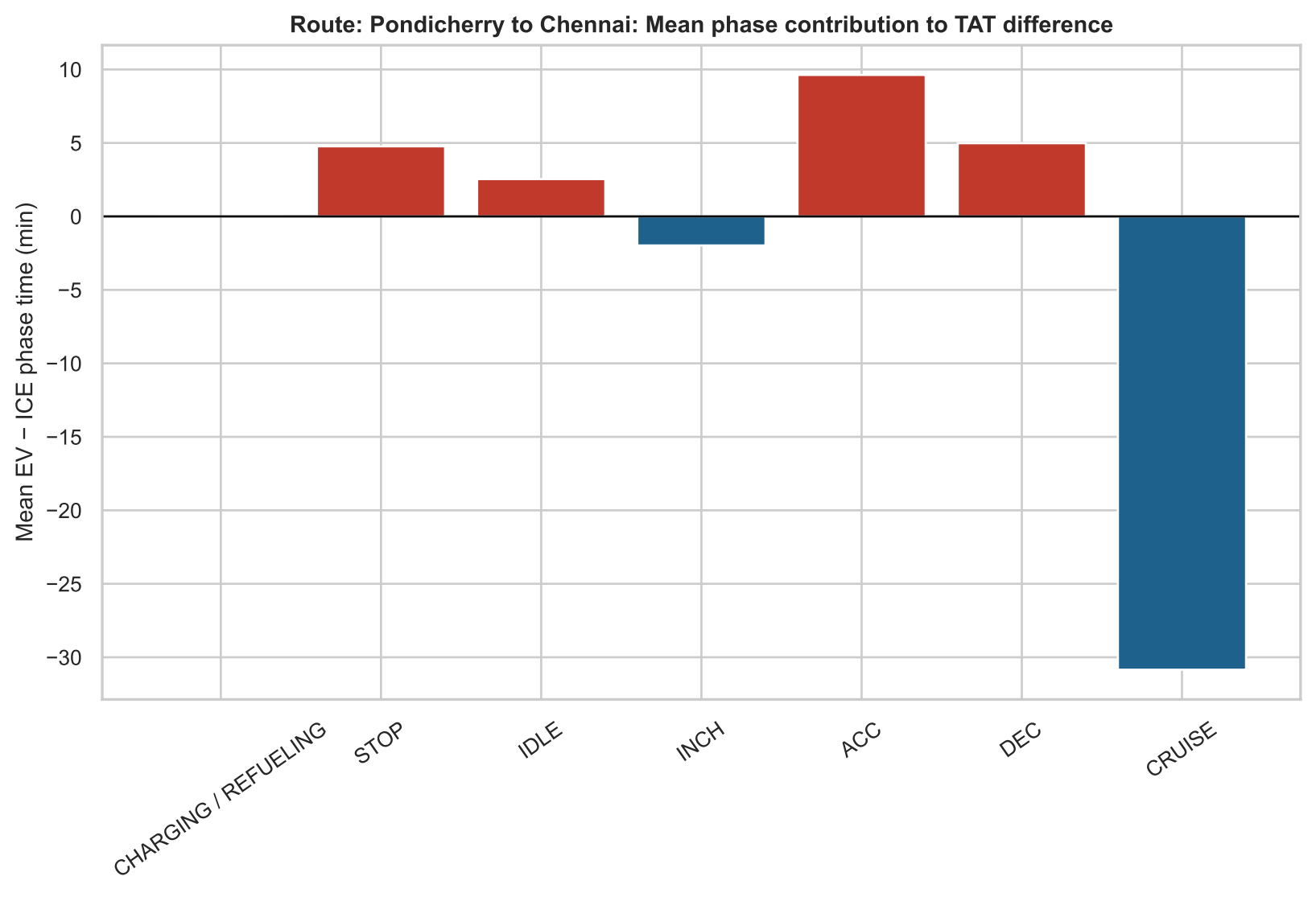
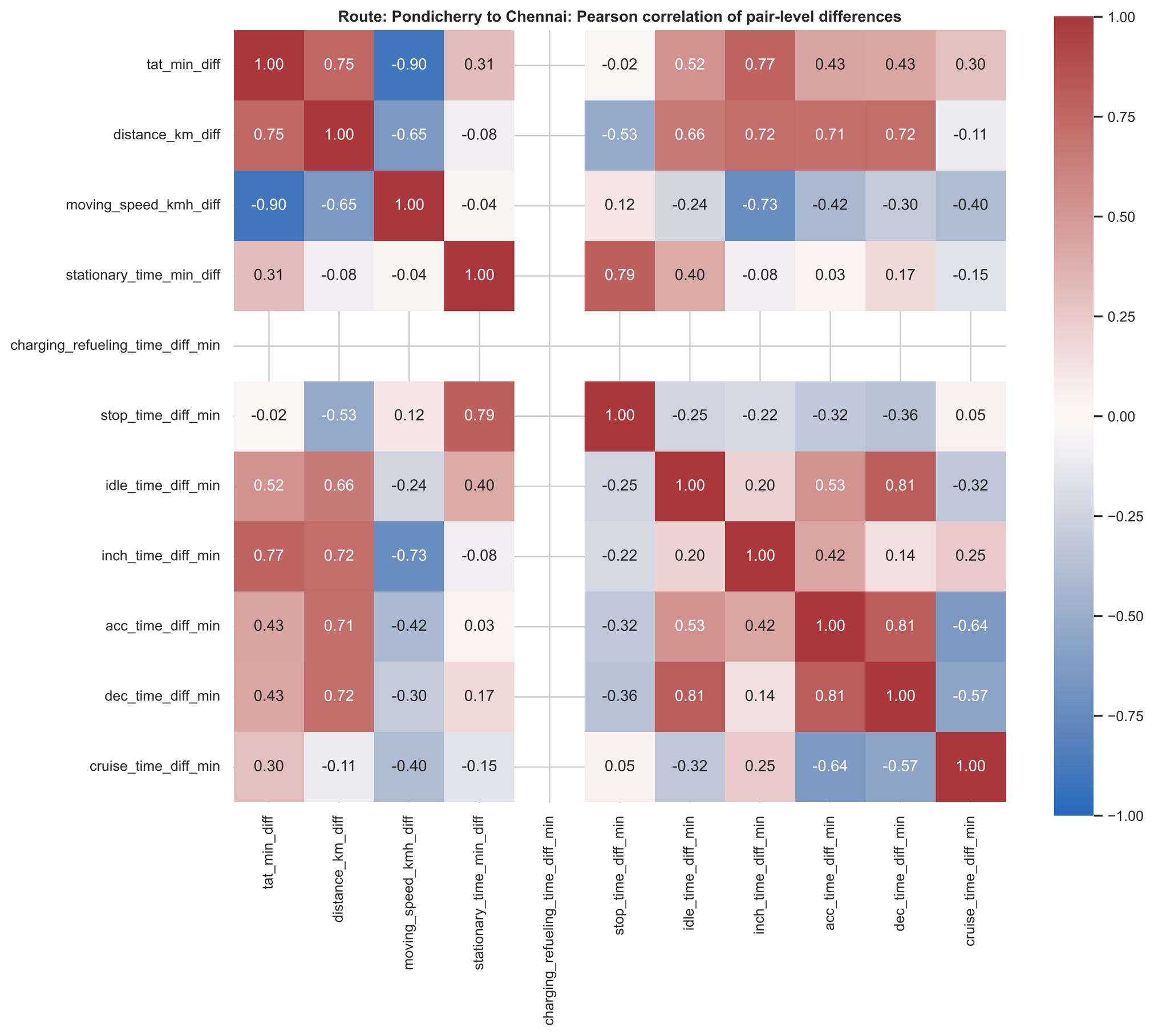

Route: Pondicherry to Chennai: Ranked pair-level TAT difference
PAIR_024
PAIR_014
PAIR_013
PAIR_017
PAIR_035
PAIR_011
PAIR_031
PAIR_023
PAIR_026
PAIR_027
PAIR_016
PAIR_015
PAIR_052
PAIR_012
PAIR_034
PAIR_025
−40
−30
−20
−10
0
10
EV TAT − ICE TAT (min)

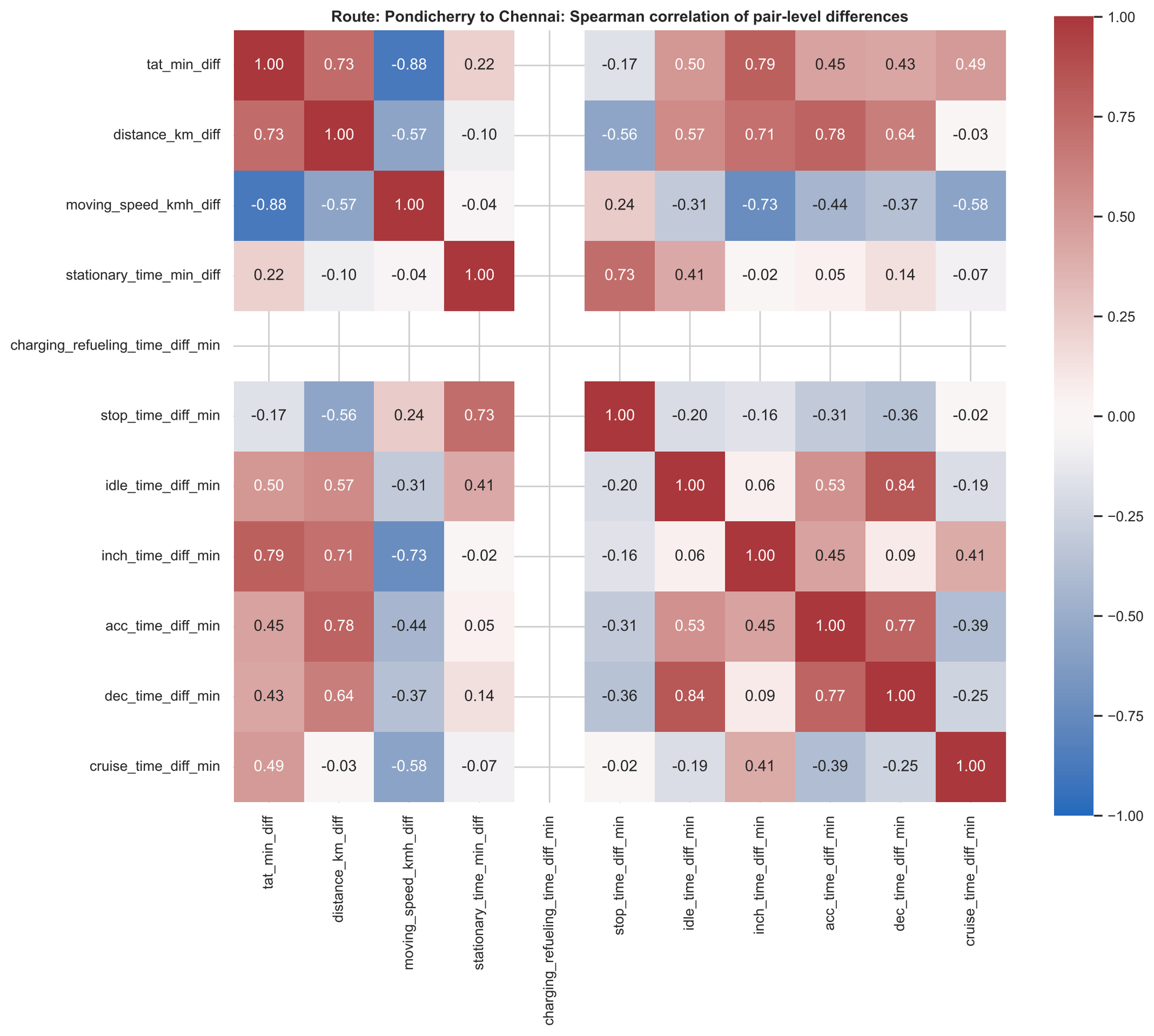

Route: Pondicherry to Chennai: Distribution of TAT differences
Count
0
1
2
3
4
5
−40
−30
−20
−10
0
10
EV TAT − ICE TAT (min)

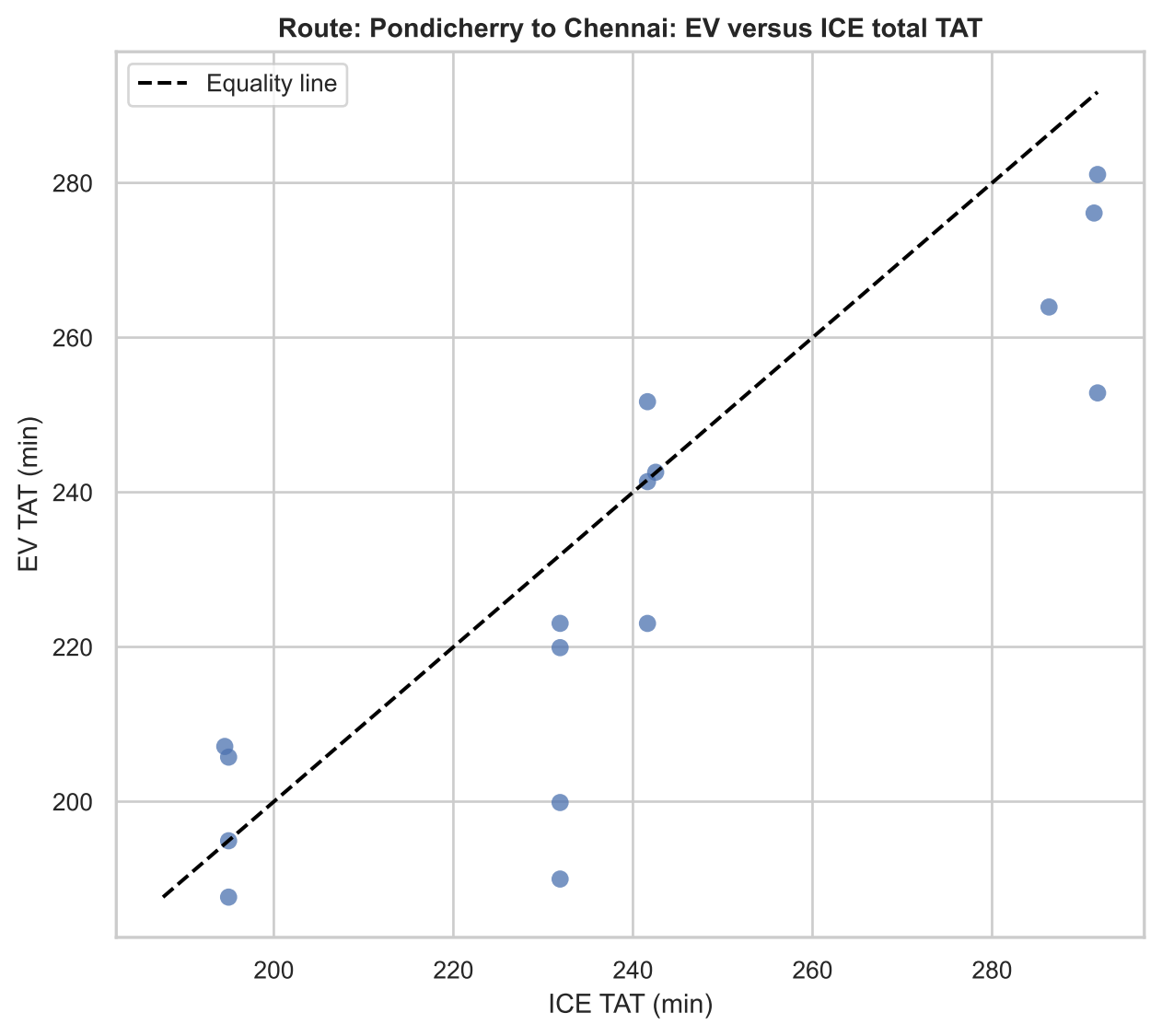

Route: Pondicherry to Chennai: EV and ICE TAT distribution
Total TAT (min)
280
260
240
220
200
EV
ICE

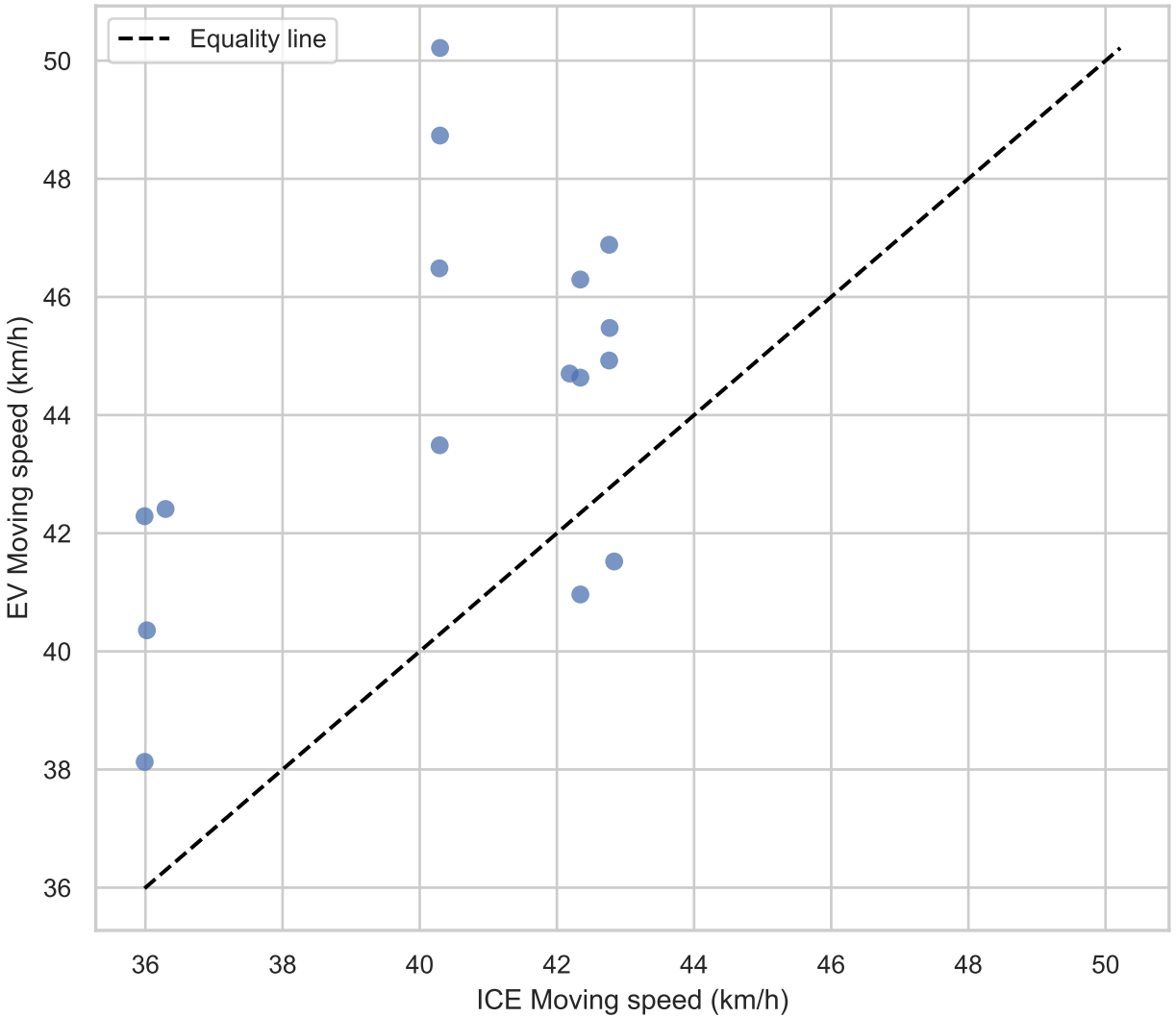

Route: Pondicherry to Chennai: Pair-by-phase time difference
pair_id
PAIR_025
PAIR_034
PAIR_012
PAIR_052
PAIR_015
PAIR_016
PAIR_027
PAIR_026
PAIR_023
PAIR_031
PAIR_011
PAIR_035
PAIR_017
PAIR_013
PAIR_014
PAIR_024
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
30
20
10
0
−10
−20
−30

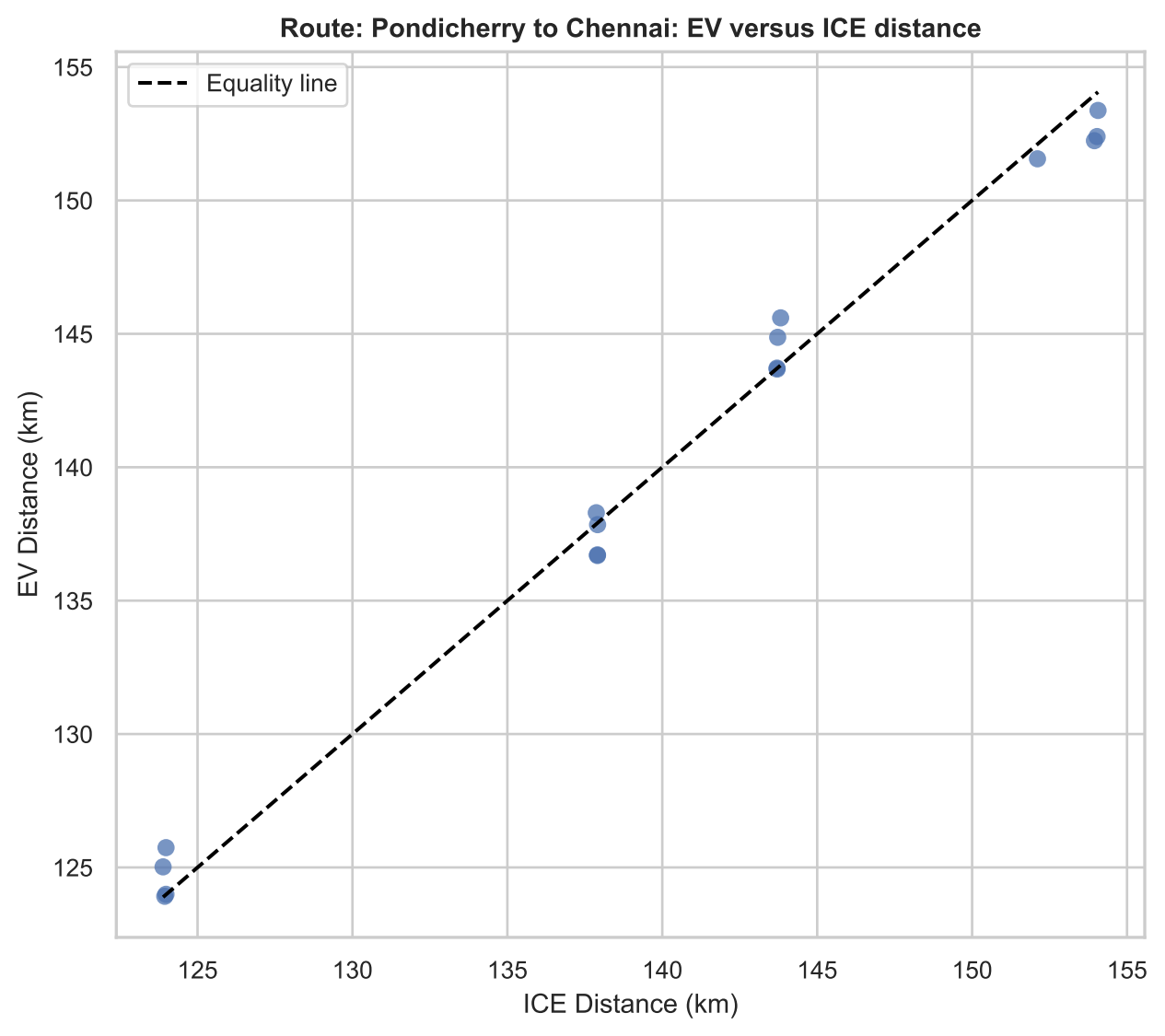

Route: Pondicherry to Chennai: Mean phase contribution to TAT difference
Mean EV − ICE phase time (min)
10
5
0
−5
−10
−15
−20
−25
−30
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

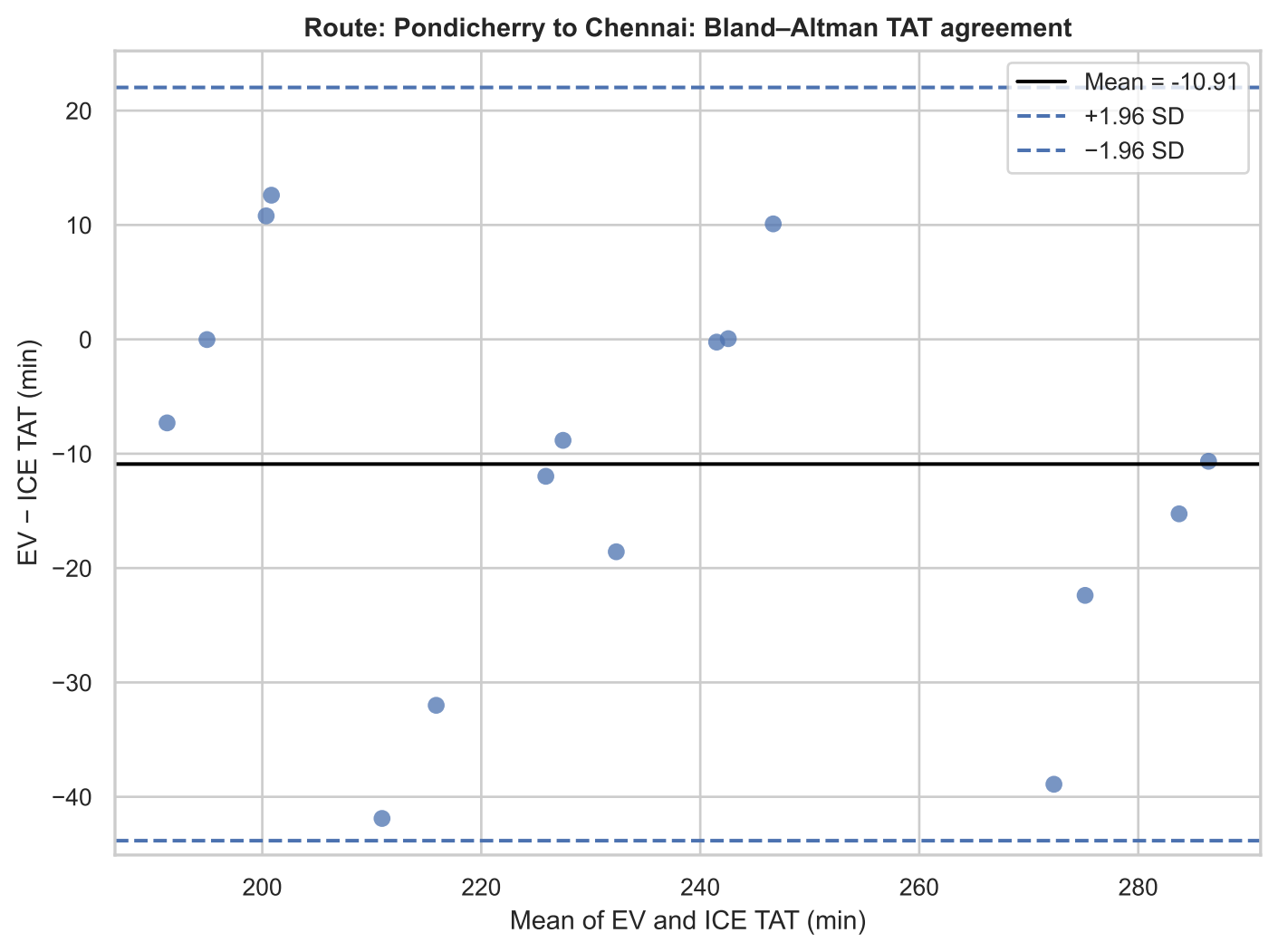

Route: Pondicherry to Chennai: Pearson correlation of pair-level differences
tat_min_diff 1.00 0.75 -0.90 0.31 -0.02 0.52 0.77 0.43 0.43 0.30
distance_km_diff 0.75 1.00 -0.65 -0.08 -0.53 0.66 0.72 0.71 0.72 -0.11
moving_speed_kmh_diff -0.90 -0.65 1.00 -0.04 0.12 -0.24 -0.73 -0.42 -0.30 -0.40
stationary_time_min_diff 0.31 -0.08 -0.04 1.00 0.79 0.40 -0.08 0.03 0.17 -0.15
charging_refueling_time_diff_min
stop_time_diff_min -0.02 -0.53 0.12 0.79 1.00 -0.25 -0.22 -0.32 -0.36 0.05
idle_time_diff_min 0.52 0.66 -0.24 0.40 -0.25 1.00 0.20 0.53 0.81 -0.32
inch_time_diff_min 0.77 0.72 -0.73 -0.08 -0.22 0.20 1.00 0.42 0.14 0.25
acc_time_diff_min 0.43 0.71 -0.42 0.03 -0.32 0.53 0.42 1.00 0.81 -0.64
dec_time_diff_min 0.43 0.72 -0.30 0.17 -0.36 0.81 0.14 0.81 1.00 -0.57
cruise_time_diff_min 0.30 -0.11 -0.40 -0.15 0.05 -0.32 0.25 -0.64 -0.57 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

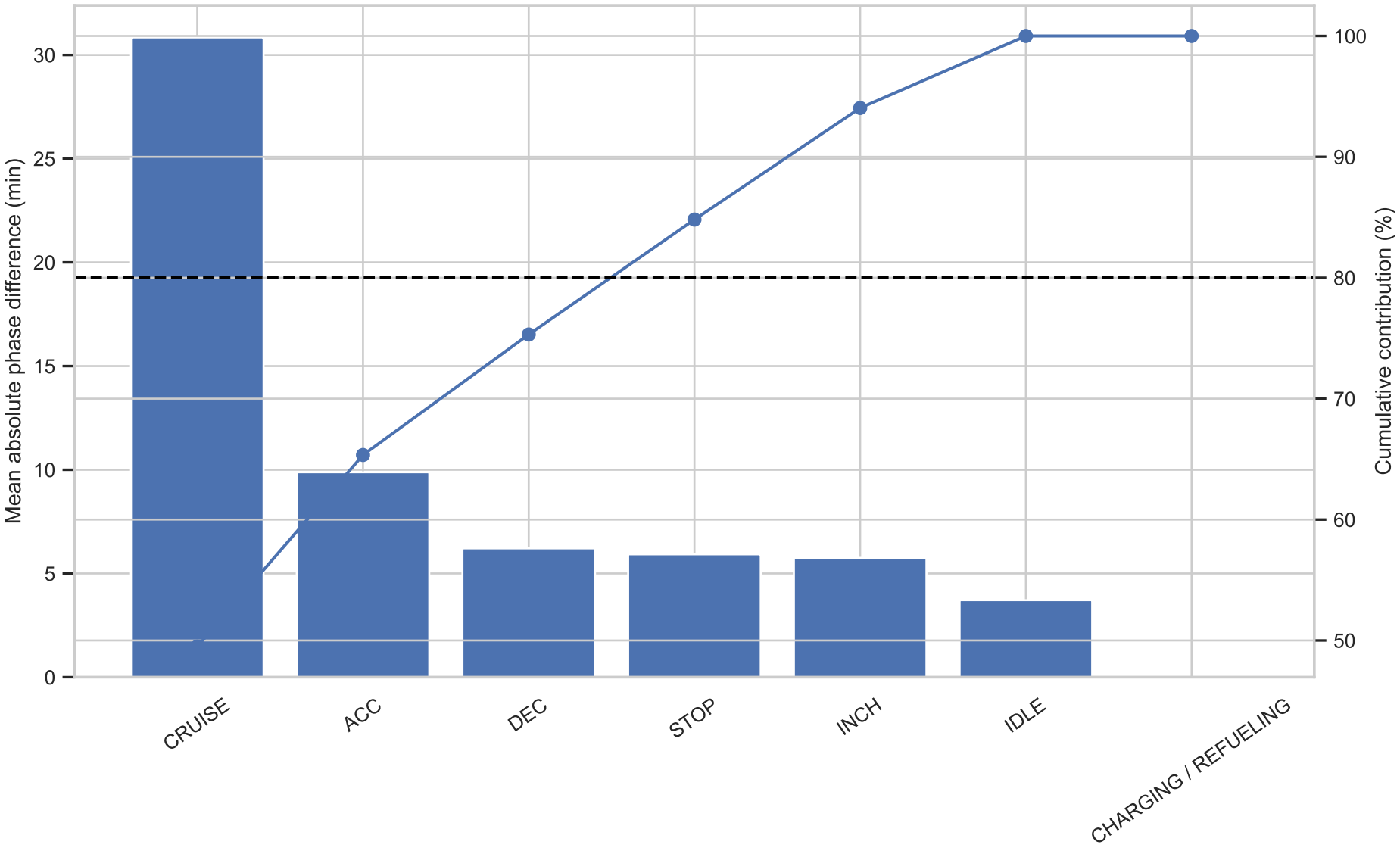

Route: Pondicherry to Chennai: Spearman correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00 0.73 -0.88 0.22 -0.17 0.50 0.79 0.45 0.43 0.49
0.73 1.00 -0.57 -0.10 -0.56 0.57 0.71 0.78 0.64 -0.03
-0.88 -0.57 1.00 -0.04 0.24 -0.31 -0.73 -0.44 -0.37 -0.58
0.22 -0.10 -0.04 1.00 0.73 0.41 -0.02 0.05 0.14 -0.07
-0.17 -0.56 0.24 0.73 1.00 -0.20 -0.16 -0.31 -0.36 -0.02
0.50 0.57 -0.31 0.41 -0.20 1.00 0.06 0.53 0.84 -0.19
0.79 0.71 -0.73 -0.02 -0.16 0.06 1.00 0.45 0.09 0.41
0.45 0.78 -0.44 0.05 -0.31 0.53 0.45 1.00 0.77 -0.39
0.43 0.64 -0.37 0.14 -0.36 0.84 0.09 0.77 1.00 -0.25
0.49 -0.03 -0.58 -0.07 -0.02 -0.19 0.41 -0.39 -0.25 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

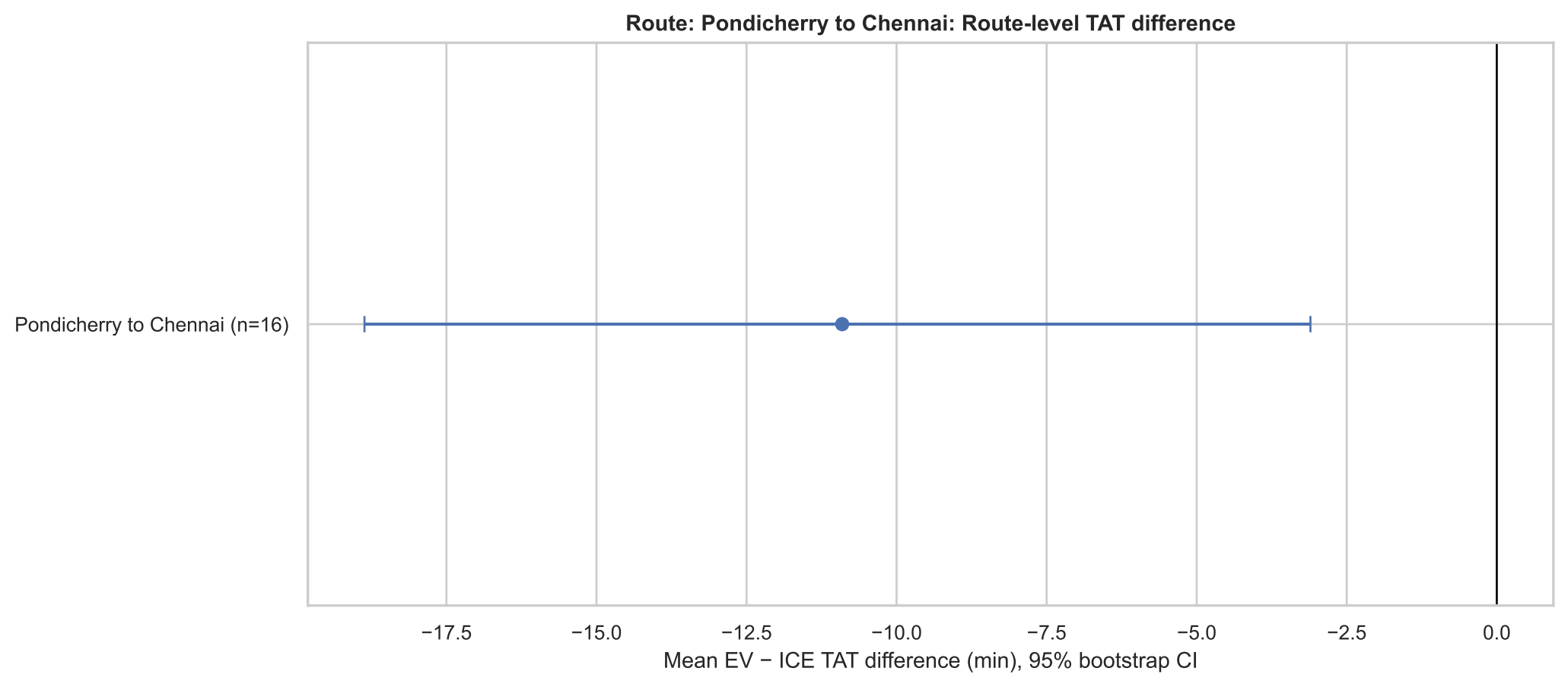

Route: Pondicherry to Chennai: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
200
220
240
260
280

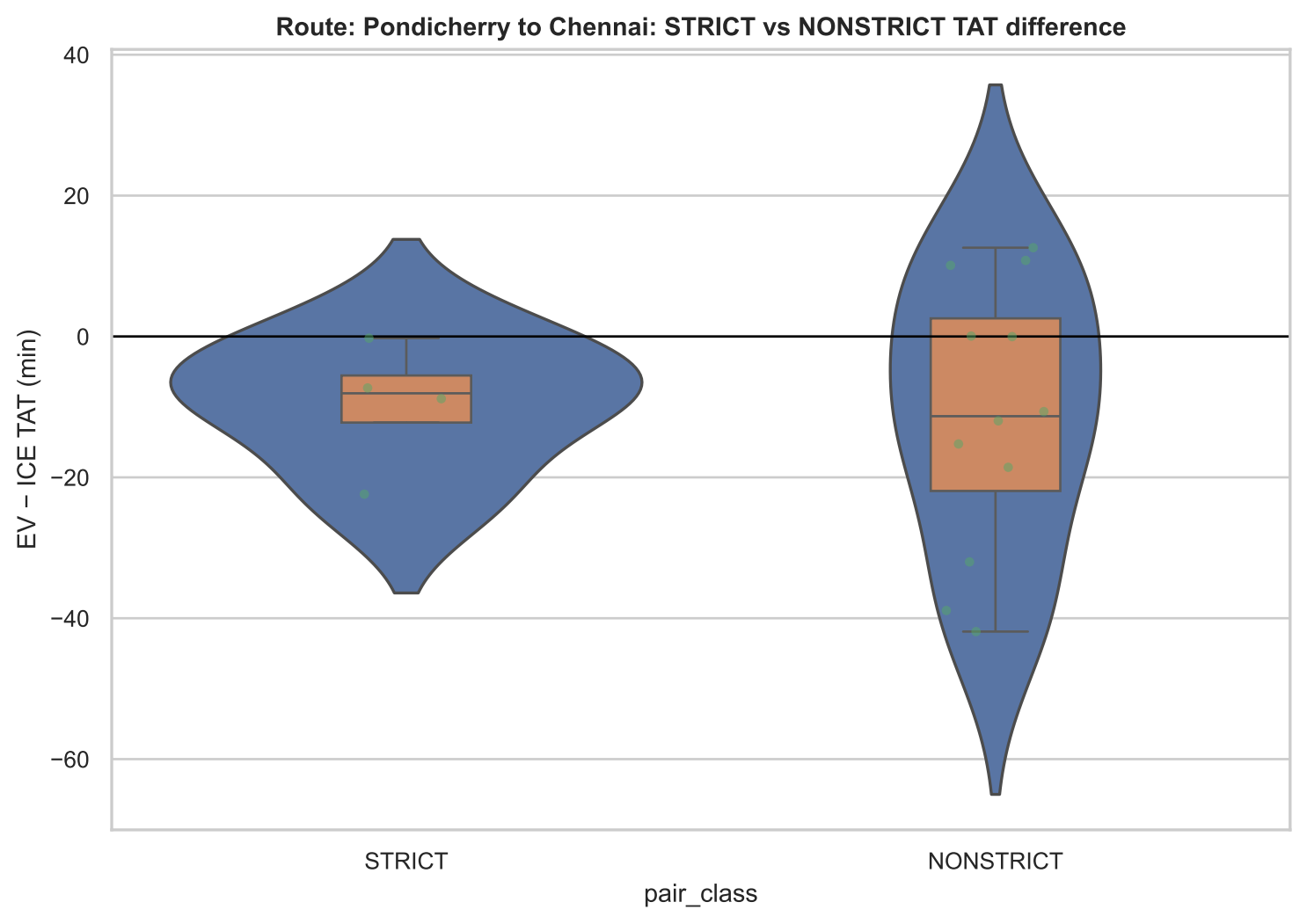

Route: Pondicherry to Chennai: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
36
38
40
42
44
46
48
50

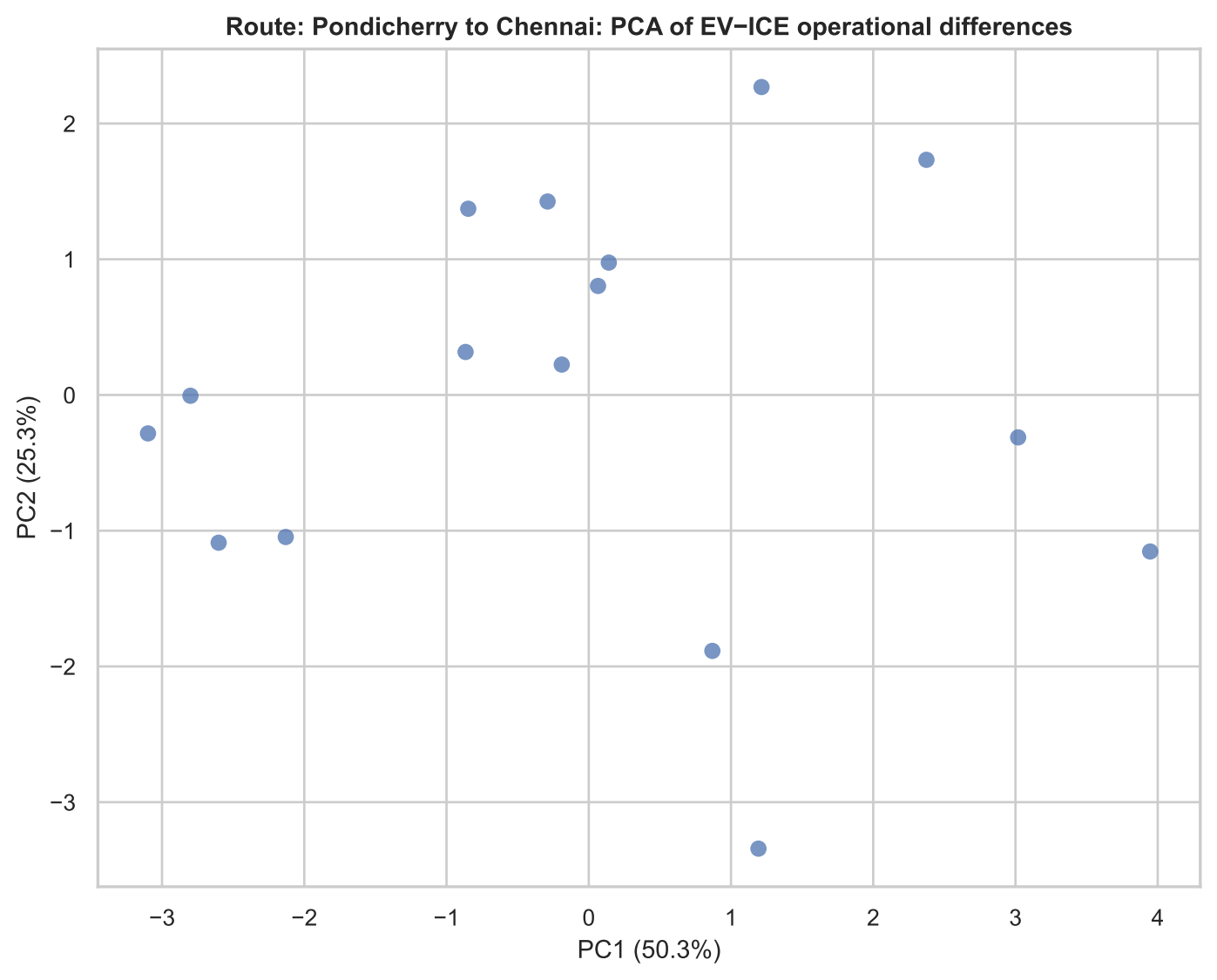

Route: Pondicherry to Chennai: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
155
150
145
140
135
130
125
125
130
135
140
145
150
155

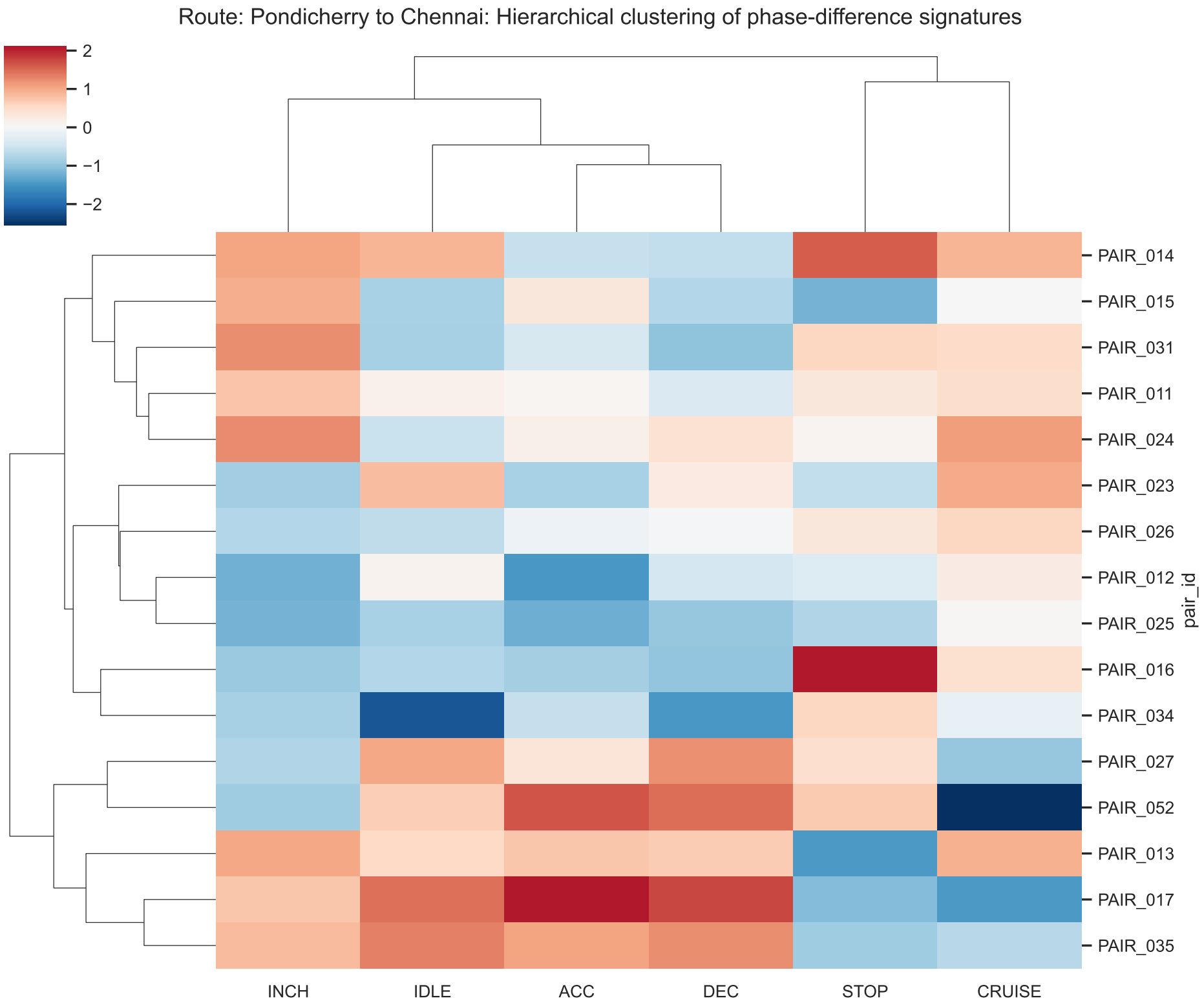

Route: Pondicherry to Chennai: Bland–Altman TAT agreement
Mean = -10.91
+1.96 SD
−1.96 SD
20
10
0
−10
−20
−30
−40
EV − ICE TAT (min)
200
220
240
260
280
Mean of EV and ICE TAT (min)


Route: Pondicherry to Chennai: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
5
10
15
20
25
30
50
60
70
80
90
100
CRUISE
ACC
DEC
STOP
INCH
IDLE
CHARGING / REFUELING


Route: Pondicherry to Chennai: Route-level TAT difference
Pondicherry to Chennai (n=16)
−17.5
−15.0
−12.5
−10.0
−7.5
−5.0
−2.5
0.0
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Pondicherry to Chennai: STRICT vs NONSTRICT TAT difference
40
20
0
−20
−40
−60
EV – ICE TAT (min)
STRICT
NONSTRICT
pair_class


Route: Pondicherry to Chennai: PCA of EV−ICE operational differences
PC2 (25.3%)
PC1 (50.3%)
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3
4


Route: Pondicherry to Chennai: Hierarchical clustering of phase-difference signatures
2
1
0
−1
−2
PAIR_014
PAIR_015
PAIR_031
PAIR_011
PAIR_024
PAIR_023
PAIR_026
PAIR_012
PAIR_025
PAIR_016
PAIR_034
PAIR_027
PAIR_052
PAIR_013
PAIR_017
PAIR_035
pair_id
INCH
IDLE
ACC
DEC
STOP
CRUISE